

EE 446: TinyML

Lab 8: TinyML for Motion Classification – On Device Data Collection & Ensemble Learning

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Today's Lab

Part I (Edge Impulse): On Device Data Collection and Motion Classification

- Both Software and Hardware Lab!
- Accelerometer Data Collection for Motion Classification
- Computing FFT, Spectrogram, and Power Spectral Density (PSD)
- Feature Extraction from Time Series IMU Data
- Training ML Models for Motion Classification
- Implementing the Model on Arduino Nano 33 BLE Sense using TinyML



Lab 8 – Part I - Software - Road Map

1. Installing Arduino CLI
2. Installing Edge Impulse Firmware on Arduino
3. Data Collection and Labelling
4. Splitting Data
5. Spectral Analysis
6. Training the ML model (DNN)



Lab 8 – Part I - Software - Road Map

7. Evaluation of ML Model Performance

8. Building ML model



Lab 8 – Part I - Software Lab – Arduino CLI Installation

If you're on Windows:

- Unzip the .zip file to C:\Program Files\arduino-cli
- Open System Properties > Advanced > Environment Variables
- Path under "user variables" > Edit
- Add C:\Program Files\arduino-cli

You can skip this step if you have already installed Arduino CLI in Previous Labs

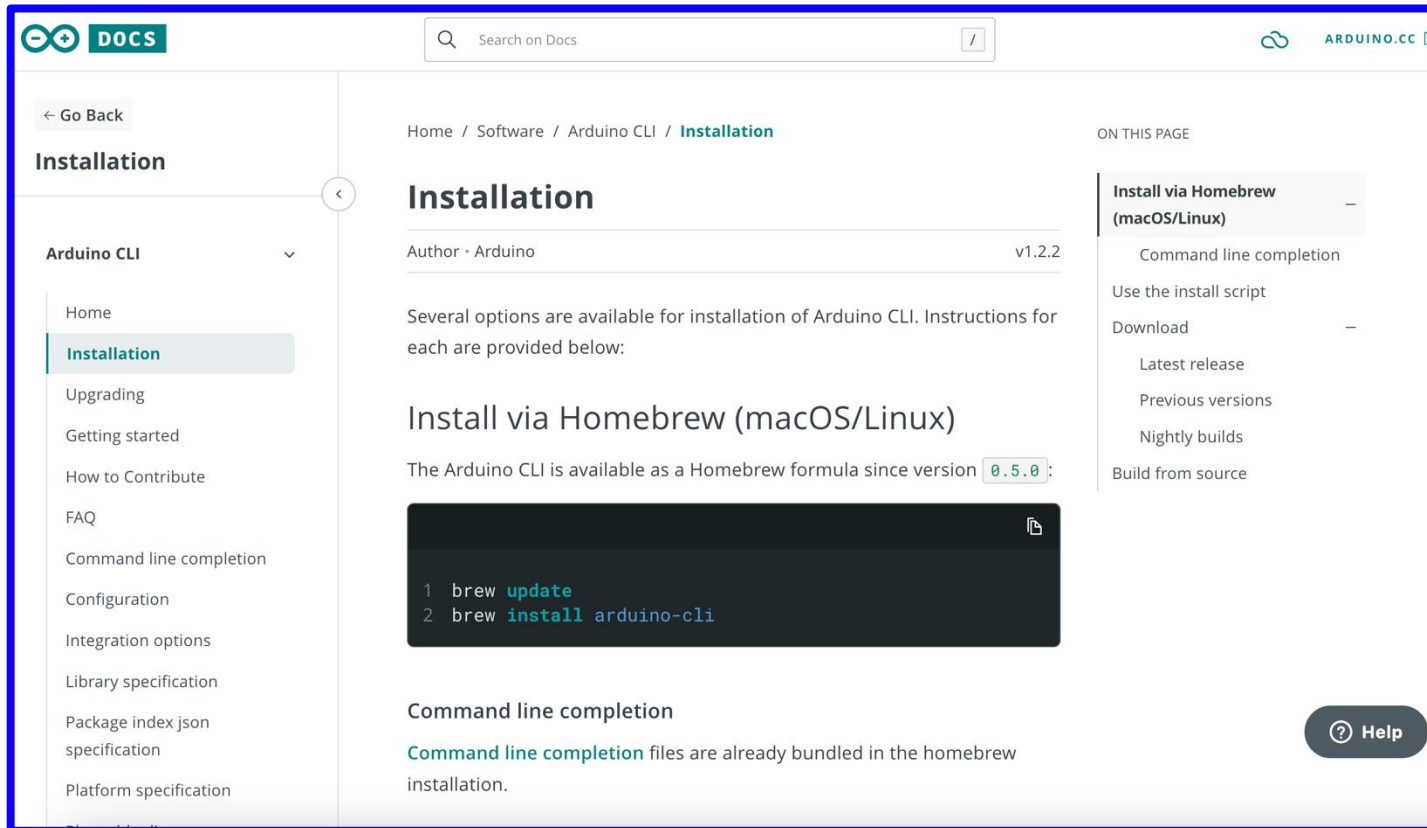
If you're on macOS:

- Use the curl method or homebrew shown on the installation page:
<https://arduino.github.io/arduino-cli/0.21/installation/>
- If you use the curl method, it likely installs arduino-cli to ~/bin. That might not be on your path. So, you might need to run: export PATH=\$PATH:~/bin



Lab 8 – Part I – Software Lab: Setting up Arduino CLI

- Go to <https://docs.arduino.cc/arduino-cli/installation/>
- Carefully follow instructions corresponding to your Laptop's/PC's operating system (Windows, MAC, Linux) for Download and Install Arduino CLI



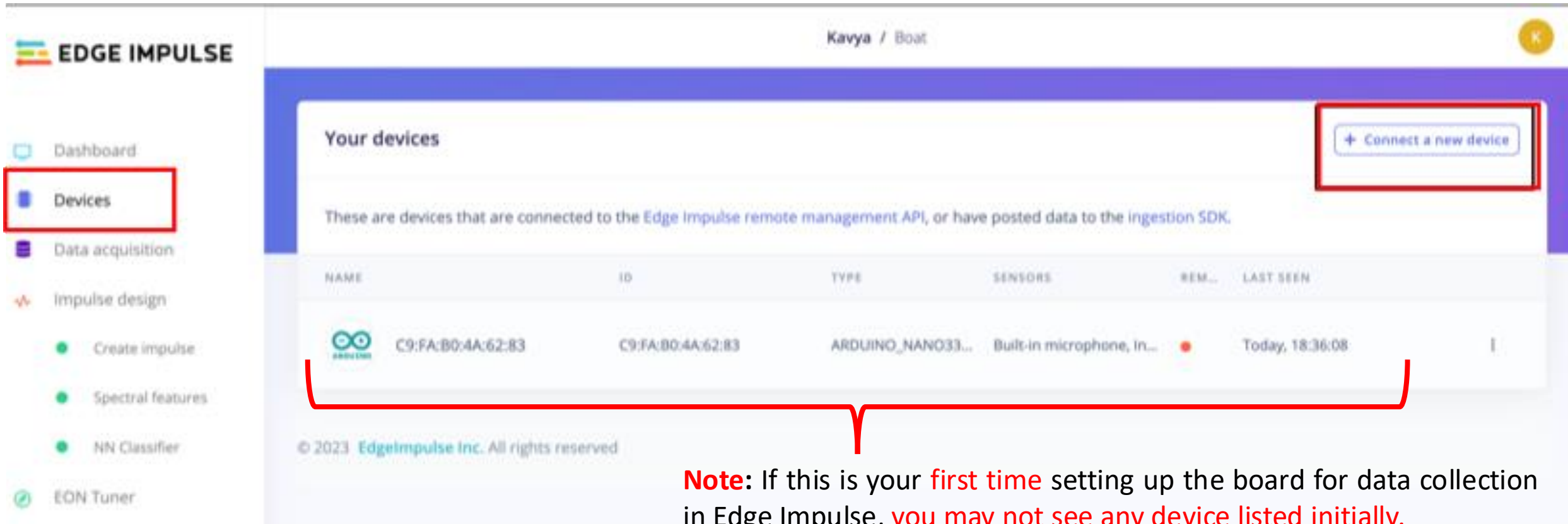
You can skip this step if you have already installed Arduino CLI in Previous Labs

Lab 8 – Part I – Software Lab – Edge Impulse Project Creation

- We will use Edge Impulse for this Lab.
Website: <https://www.edgeimpulse.com/>
(You can login in using the account we created for the previous Edge Impulse lab)
- Create a new Project on Edge Impulse and provide the title as “Motion Classification for Transport”.

Software Lab – Connecting the Device

- First connect your Arduino Nano BLE to your Laptop/PC using USB C Cable
- Then in your Edge Impulse Project go to “Devices”



The screenshot shows the Edge Impulse web interface. On the left sidebar, the 'Devices' menu item is highlighted with a red box. In the main content area, the 'Your devices' section is visible. A red box highlights the '+ Connect a new device' button in the top right corner of this section. Below the button, there is a table of connected devices. A red bracket is drawn under the table.

NAME	ID	TYPE	SENSORS	REM...	LAST SEEN
 C9:FA:B0:4A:62:83	C9:FA:B0:4A:62:83	ARDUINO_NANO33...	Built-in microphone, In...		Today, 18:36:08

Note: If this is your **first time** setting up the board for data collection in Edge Impulse, **you may not see any device listed initially.**

Software Lab - Connecting the Device

Collect new data

Collect data directly from your phone, computer, device, or development board.



Scan QR code to connect to your phone



Connect to your computer



Connect your device or development board

Once you select **“Arduino Nano 33 BLE Sense,”** Edge Impulse will open a setup page with the required installation steps.

If you have already successfully installed both CLIs — **Edge Impulse CLI** and **Arduino CLI** — you can skip the CLI installation steps and **continue with the following steps on that page:**

- 1. Connect the development board to your computer.**
- 2. Update/flash the firmware on the board.**

Connect your device or development board

You can connect almost any device over serial using our [data forwarder](#):

```
$ edge-impulse-data-forwarder
```

Or connect a supported [device](#) or [development board](#):

Search supported development boards



Any device running macOS



Arducam Pico4ML TinyML Dev Kit



Arduino Nano 33 BLE Sense



Arduino Nicla Vision

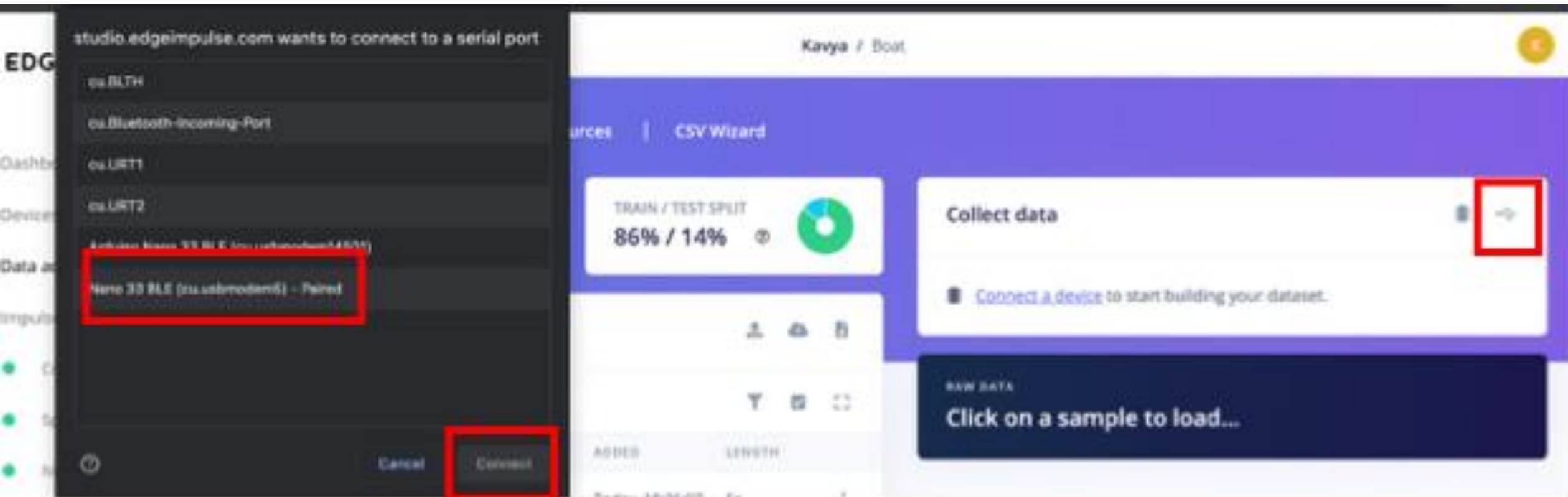


Arduino Nicla Voice



Software Lab – Data Acquisition

- ❖ Go to the **Data Acquisition** tab in the **left-hand pane** of your Edge Impulse project.



Software Lab – Data Acquisition

The screenshot shows the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Impulse design, Create impulse, EON Tuner, Retrain model, Live classification, Model testing, Versioning, and Deployment. The main area has tabs for Dataset, Data explorer, Data sources, and CSV Wizard. The Dataset tab is active, showing a summary of data collected (1m 45s) and a train/test split (86% / 14%). Below this is a table of the dataset with columns for Sample Name, Label, Added, and Length. The table lists 11 samples, all labeled 'idle', with lengths of 5s or 3s. To the right of the table is the 'Collect data' form. It includes a 'Device' dropdown (currently 'No devices connected'), a 'Label' input field (containing 'lift'), a 'Sample length (ms.)' input field (containing '10000'), a 'Sensor' dropdown, and a 'Frequency' dropdown. A 'Start sampling' button is at the bottom of the form. Below the form is a 'RAW DATA' section showing a waveform plot for a specific sample.

EDGE IMPULSE

Kavya / Boat

Dataset | Data explorer | Data sources | CSV Wizard

DATA COLLECTED
1m 45s

TRAIN / TEST SPLIT
86% / 14%

Dataset

Training (18) | Test (3)

SAMPLE NAME	LABEL	ADDED	LENGTH
idle.3vu8ps33.inge...	idle	Today, 18:36:07	5s
idle.3vu8ps33.inge...	idle	Today, 18:36:07	5s
idle.3vu8ps33.inge...	idle	Today, 18:36:06	5s
idle.3vu8ps33.inge...	idle	Today, 18:36:06	5s
idle.3vu8ps33.inge...	idle	Today, 18:36:06	5s
idle.3vu8ps33.inge...	idle	Today, 18:36:06	5s
idle.3vu8ps33.s8	idle	Today, 18:26:02	3s
idle.3vu8ps33.s7	idle	Today, 18:26:02	5s
idle.3vu8ps33.s6	idle	Today, 18:26:02	5s

Collect data

Device ?
No devices connected

Label
lift

Sample length (ms.)
10000

Sensor

Frequency

Start sampling

RAW DATA
idle.3vu8ps33.ingestion-7f6f59c885-zgppq.s6

For Lab 8, create 20 samples per class— 'Lift' and 'Idle'— each sample lasting 10 seconds.

2. Add you Label

1. Select
Appropriate
Sensors (IMU)

3. Press to Collect
Data point



Software Lab – Impulse Design – Create Impulse



Dashboard

Devices

Data acquisition

Impulse design

- Create impulse
- Spectral features
- NN Classifier

EON Tuner

Retrain model

Live classification

Model testing

Versioning

Deployment

An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.

Time series data

Input axes (9)
accX, accY, accZ, gyrX, gyrY, gyrZ, magX, magY, magZ

Window size
2000 ms.

Window increase
80 ms.

Frequency (Hz)
50

Zero-pad data
☒

Spectral Analysis

Name
Spectral features

Input axes (3)

- ☒ accX
- ☒ accY
- ☒ accZ
- ☐ gyrX
- ☐ gyrY
- ☐ gyrZ
- ☐ magX
- ☐ magY
- ☐ magZ

Classification

Name
NN Classifier

Input features
☒ Spectral features

Output features
2 (idle, lift)

Add a learning block

Output features

2 (idle, lift)

Save Impulse

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Software Lab – Impulse Design – Spectral Features

Raw features 

0.4300, 0.9400, 11.3200, 0.1300, 0.4300, 11.7400, -0.1200, 0.4700, 11.0600, -0.0900, 0.7600,...

Parameters

Filter

Scale axes

Input decimation ratio

Type

Cut-off frequency

Order

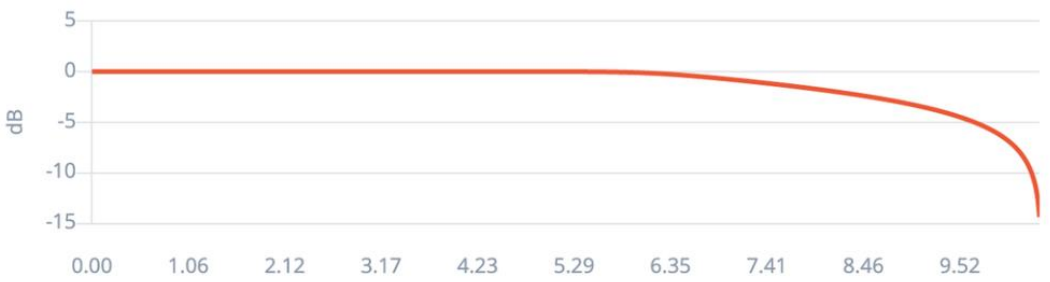
Analysis

Type

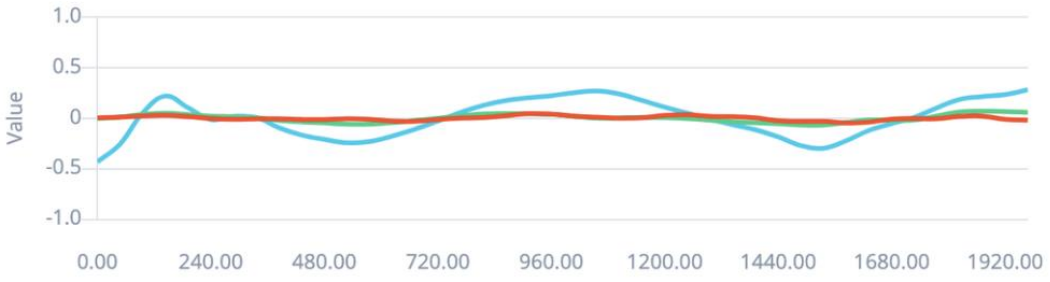
Autotune parameters

DSP result

Filter response



After filter



Spectral power (log)



[illegible]

The screenshot displays the Google Cloud AI Platform interface for feature generation. At the top, there's a header bar with a dropdown menu set to '#1' and a link to 'Click to set a description for this version'. Below this, the 'Parameters' section includes a red-bordered box around the 'Generate features' button. The main area is divided into three panels:

- Training set**: Shows metadata such as 'Data in training set' (14m 32s), 'Classes' (4: idle, snake, updown, wave), 'Training windows' (3,600), and a checkbox for 'Calculate feature importance'.
- Feature explorer**: Contains a scatter plot visualizing four classes: idle (blue), snake (orange), updown (green), and wave (red). A specific window labeled 'updown.5 (5376 - 7376 ms.)' is highlighted. Below the plot, a time-series graph shows acceleration data (accX, accY, accZ) over time (0ms to 8703ms).
- Feature generation output**: Displays a log of messages indicating the progress of embedding generation, ending with 'Job completed'.

A red bracket on the right side of the Feature Explorer panel points to the scatter plot with the annotation: "You should see only two classes here (idle and lift)".

Software Lab – Impulse Design - NN Classifier

The screenshot displays the Edge Impulse web interface for a project titled "Motion Classification for Transport". The left sidebar contains navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Spectral features, Classifier, Retrain model, Live classification, Model testing, Deployment, Versioning, Documentation, and an Upgrade Plan. The main content area is divided into two panels. The left panel, titled "Neural Network settings", is highlighted with a red border and contains three sections: "Training settings" with fields for "Number of training cycles" (30), "Use learned optimizer" (unchecked), "Learning rate" (0.1), and "Training processor" (CPU); "Advanced training settings" (collapsed); and "Neural network architecture" showing a sequence of layers: "Input layer (39 features)", "Dense layer (20 neurons)", "Dense layer (10 neurons)", "Add an extra layer" (dashed box), and "Output layer (2 classes)". A "Save & train" button is at the bottom right of this panel. The right panel, titled "Training output", shows a log of training jobs. The first job is completed, showing a success message and a list of steps: "Job scheduled at 13 May 2025 22:59:51", "Job started at 13 May 2025 22:59:53", "Copying features from processing blocks...", "Copying features from DSP block...", "Copying features from DSP block OK", and "Copying features from processing blocks OK". The second job is also completed, showing a success message and steps: "Job scheduled at 13 May 2025 23:00:02", "Job started at 13 May 2025 23:00:03", "Splitting data into training and validation sets...", and "Splitting data into training and validation sets OK". Below the training output, the "Model" section shows the "Model version" as "Quantized (int8)". The "Last training performance (validation set)" section displays "ACCURACY" as 84.8% and "LOSS" as 0.41. The "Confusion matrix (validation set)" section shows a table with rows for "IDLE", "LIFT", and "F1 SCORE" and columns for "IDLE" and "LIFT". The "Metrics (validation set)" section shows a table with rows for "Area under ROC Curve", "Weighted average Precision", "Weighted average Recall", and "Weighted average F1 score", all with a value of 0.85.

EDGE IMPULSE

Dashboard
Devices
Data acquisition
Experiments
EON Tuner
Impulse design
Create impulse
Spectral features
Classifier
Retrain model
Live classification
Model testing
Deployment
Versioning
Documentation
Upgrade Plan
Get access to higher job limits and more collaborators.
View plans

eychloe / Motion Classification for Transport PERSONAL Target: Arduino Nano 33 ...

Neural Network settings

Training settings

Number of training cycles ② 30

Use learned optimizer ② ☐

Learning rate ② 0.1

Training processor ② CPU

Advanced training settings

Neural network architecture

Input layer (39 features)

Dense layer (20 neurons)

Dense layer (10 neurons)

Add an extra layer

Output layer (2 classes)

Save & train

Training output

Creating job... OK (ID: 32943015)

✓Job scheduled at 13 May 2025 22:59:51
✓Job started at 13 May 2025 22:59:53
Copying features from processing blocks...
Copying features from DSP block...
Copying features from DSP block OK
Copying features from processing blocks OK

✓Job scheduled at 13 May 2025 23:00:02
✓Job started at 13 May 2025 23:00:03
Splitting data into training and validation sets...
Splitting data into training and validation sets OK

Model

Model version: ② Quantized (int8)

Last training performance (validation set)

ACCURACY 84.8% LOSS 0.41

Confusion matrix (validation set)

	IDLE	LIFT
IDLE	75%	25%
LIFT	5%	95%
F1 SCORE	0.83	0.86

Metrics (validation set)

METRIC	VALUE
Area under ROC Curve ②	0.85
Weighted average Precision ②	0.86
Weighted average Recall ②	0.85
Weighted average F1 score ②	0.85



Software Lab – Deployment

 **EDGE IMPULSE**

Dashboard

Devices

Data acquisition

Experiments

EON Tuner

Time series data, S...

Create impulse

Spectral features

NN Classifier

Anomaly detection

Retrain model

Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

Configure your deployment

You can deploy your impulse to any device. This makes the model run without an internet connection, minimizes latency, and runs with minimal power consumption. [Read more.](#)

Arduino Nano 33 BLE Sense

SELECTED DEPLOYMENT

 **Arduino Nano 33 BLE Sense**
A binary containing both the Edge Impulse data acquisition client and your full impulse.

MODEL OPTIMIZATIONS

Model optimizations can increase on-device performance but may reduce accuracy.

 **EON™ Compiler**
Same accuracy, 36% less RAM, 40% less ROM.

Quantized (int8)
Selected ✓

	SPECTRAL FEATUR...	NN CLASSIFIER	TOTAL
LATENCY	18 ms.	1 ms.	19 ms.
RAM	8.1K	1.7K	8.1K
FLASH	-	20.0K	-
ACCURACY			-

Latest build

 **v12 (Arduino Nano 33 BLE Sense)**
Today, 16:00:06 [View docs](#)

Build output

Cancel

Creating job... OK (ID: 32219443)

- ✓ Job scheduled at 29 Apr 2025 23:06:57
- ✓ Job started at 29 Apr 2025 23:07:00
- Writing templates...
- ✓ Job scheduled at 29 Apr 2025 23:07:07

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Software Lab – Model Compression

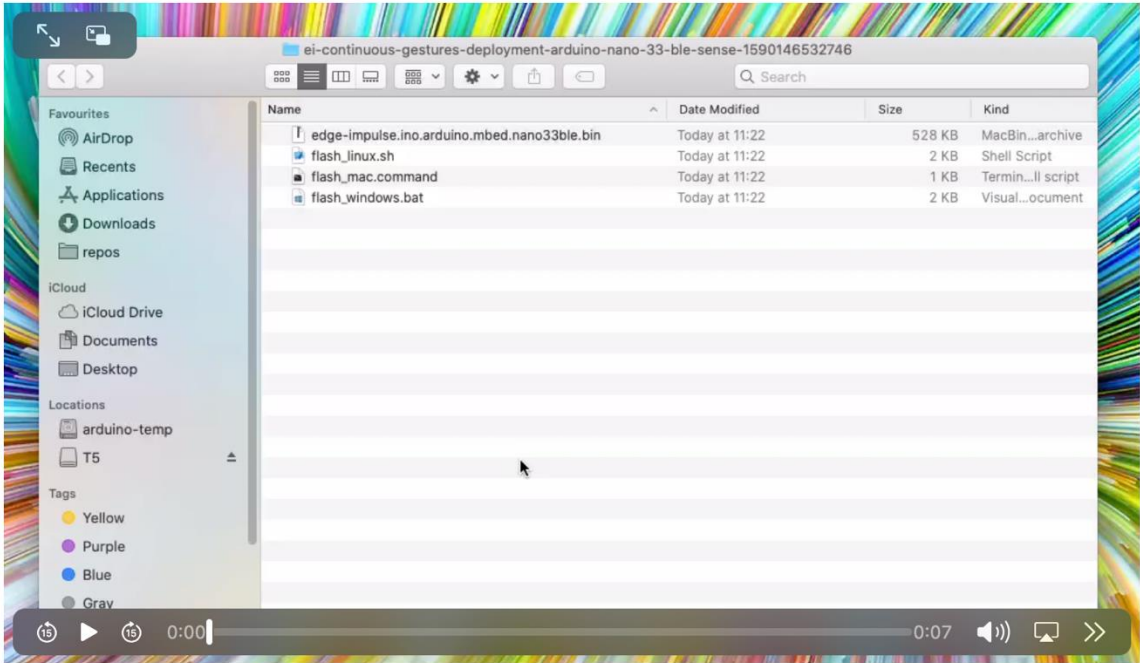
Upon a Successful
Model Build You
Should Be Able to See
Something Like This





Built Arduino Nano 33 BLE Sense firmware

Click on the script for your operating system to flash the binary.



```
bytes (27%) of program storage
83040 bytes.
e 58624 bytes (22%) of dynamic
520 bytes for local variables.
ytes.

rduino-nano-33-ble-sense done

K
```

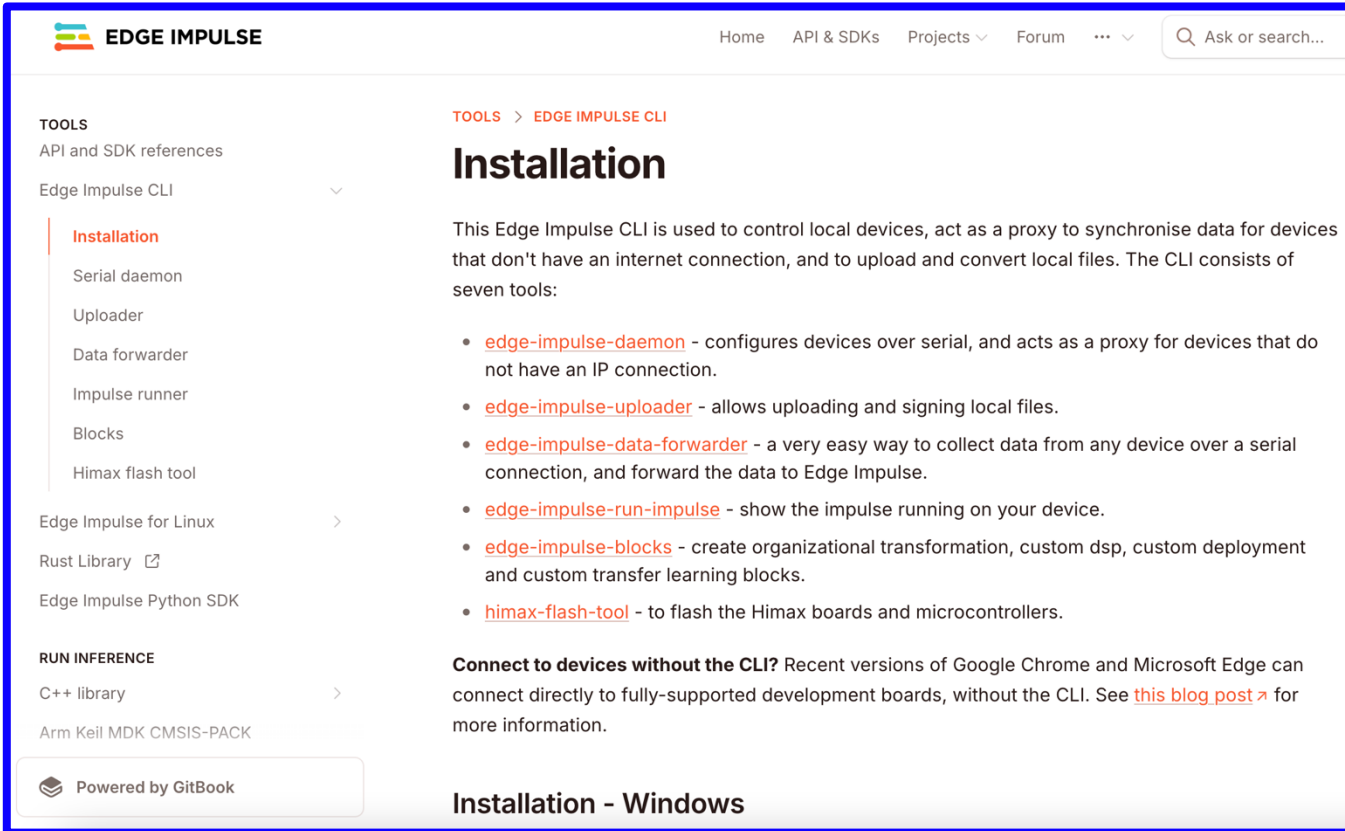
Lab 8 – Part I - Hardware - Road Map

1. Build the Arduino Library/Flash files using Edge Impulse.
2. Compile and Upload .zip/Flash the sketch to the board.
3. Open Serial Monitor/Command Prompt and test the classification.



Hardware Lab: Setting up Edge Impulse CLI

- Go to <https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-installation>
- Carefully follow instructions corresponding to your Laptop's/PC's operating system (Windows, MAC, Linux) for Download and Edge Impulse CLI

A screenshot of the Edge Impulse CLI installation page. The page has a header with the Edge Impulse logo and navigation links: Home, API & SDKs, Projects, Forum, and a search bar. The left sidebar shows a 'TOOLS' section with a dropdown menu for 'Edge Impulse CLI' containing links to 'Installation', 'Serial daemon', 'Uploader', 'Data forwarder', 'Impulse runner', 'Blocks', and 'Himax flash tool'. Below this is a 'RUN INFERENCE' section with links for 'C++ library' and 'Arm Keil MDK CMSIS-PACK'. The main content area is titled 'Installation' and explains that the CLI is used to control local devices and upload files. It lists seven tools: edge-impulse-daemon, edge-impulse-uploader, edge-impulse-data-forwarder, edge-impulse-run-impulse, edge-impulse-blocks, and himax-flash-tool. A note mentions connecting to devices without the CLI using Google Chrome or Microsoft Edge. The footer indicates the page is 'Powered by GitBook' and has a section for 'Installation - Windows'.

You can skip this step if you have already installed Edge Impulse CLI in Previous Labs

Hardware Lab – Arduino Deployment & Testing

- Flash the compressed model (downloaded from Edge Impulse) onto the board by **double clicking on the script for your operating system**
- Follow the instructions prompted in command line to observe the results in command line (*you may need to type the prompted commands in a new command prompt window*)

```
/Users/Admin/Downloads/arduino-nano-33-ble-sense\ \(\2\)flash_mac.command ; exit;  
(base) Admin@Sarats-MacBook-Pro ~ % /Users/Admin/Downloads/arduino-nano-33-ble-sense\ \(\2\)flash_mac.command ; exit  
You're using an untested version of Arduino CLI, this might cause issues (found: 0.32.2, expected: 0.18.x)  
Finding Arduino Mbed core...  
Finding Arduino Mbed OK  
Finding Arduino Nano 33 BLE...  
Finding Arduino Nano 33 BLE OK  
Flashing board...  
TOUCH: error during reset: opening port at 1200bps: Serial port busy  
Device      : nRF52840-QIAA  
Version     : Arduino Bootloader (SAM-BA extended) 2.0 [Arduino:IKXYZ]  
Address     : 0x0  
Pages      : 256  
Page Size  : 4096 bytes  
Total Size : 1024KB  
Planes     : 1  
Lock Regions : 0  
Locked     : none  
Security    : false  
Erase flash  
  
Done in 0.001 seconds  
Write 352560 bytes to flash (87 pages)  
[=====] 100% (87/87 pages)  
Done in 13.914 seconds  
  
Flashed your Arduino Nano 33 BLE development board.  
To set up your development with Edge Impulse, run 'edge-impulse-daemon'  
To run your impulse on your development board, run 'edge-impulse-run-impulse'  
  
Saving session...  
...copying shared history...  
...saving history...truncating history files...  
...completed.  
  
[Process completed]
```



Today's Lab

Part II (Google Collab + Arduino IDE): Motion Classification and Tiny Ensemble Learning

- Both Software and Hardware Lab!
- Model Ensemble Techniques for Improved Accuracy
- Compressing Models using Pruning and Quantization-Aware Training (QAT)
- Combining Multiple Classifiers with a Stacked Meta-Classifier
- Real-World Testing with Recorded IMU Data
- Deploying the Final Model on Arduino Nano 33 BLE Sense using TFLite



Lab 8 – Part II – Tiny Ensemble Learning

 Open and run [TinyML_Lab8-Part-II.ipynb](#)

 Follow the instructions and [answer the discussion questions given in TinyML_Lab8-Part-II.ipynb](#)

 **Create one PDF** that includes both **Part I** and **Part II** of Lab 8.

For **Part I**, include:

- The [public link to your Edge Impulse project](#).
- [Screenshots](#) showing that the deployed model correctly identifies [both Idle and Lift states](#).

For **Part II**, include:

- Your [completed Part II work](#), including all [required answers, screenshots, results, and explanations](#).

Upload all related files, including the [PDF, .ipynb, .ino, and any other required supporting files](#), to your [GitHub repository](#).

 Finally, submit the [GitHub repository link](#) on Canvas.

